Analog – Problem 1.1.56

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Problem Statement

Problem 1.1.56

Consider the triangular wave generator shown in Fig. 1.1.56.1. Assume ideal op-amps with $\pm 2\,\mathrm{V}$ supply. The input is a $+5\,\mathrm{V}$, 50 Hz square wave with 50% duty cycle.

The condition that results in a triangular wave of 5 V peak-to-peak and 50 Hz at the output is:

1. $RC = 1$
2. $R/C = 1$
3. $R/C = 5$
4. $C/R = 5$

Circuit Diagram

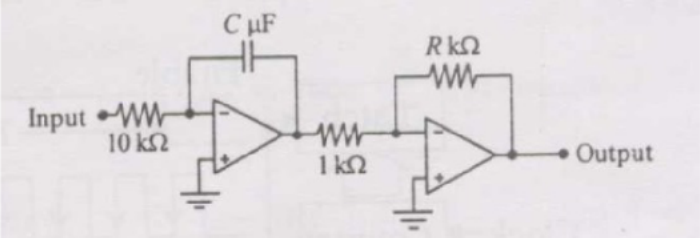

Fig : Circuit

Step 1: Deriving Output of Integrator Circuit

For an ideal op-amp with negative feedback:

1. The open-loop gain $A \to \infty$
2. The differential input voltage $V^+ - V^- \to 0$
3. Since $V^+ = 0$ (grounded), we get $V^- = 0$

The inverting terminal is therefore at **virtual ground**:

$$V^- = V^+ = 0\,\mathrm{V}$$

This means the voltage across R_1 is simply $V_{\mathrm{in}} - 0 = V_{\mathrm{in}}$.
The current through R_1 flowing into the virtual ground node is:

$$i_{R_1}(t) = \frac{V_{\mathrm{in}}(t) - V^-}{R_1} = \frac{V_{\mathrm{in}}(t)}{R_1}$$

Step 2: KCL at the Inverting Terminal

Since the ideal op-amp draws **zero input current**, all of i_{R_1} must flow through the capacitor C.
Applying KCL at the inverting node V^-:

$$i_C(t) = \frac{V_{\mathrm{in}}(t)}{R_1}$$

The voltage–current relationship for the capacitor is:

$$i_C(t) = C\,\frac{d}{dt}\big(V^- - V_{\mathrm{out}}(t)\big) = C\,\frac{d}{dt}\big(0 - V_{\mathrm{out}}(t)\big) = -C\,\frac{dV_{\mathrm{out}}(t)}{dt}$$

Step 3: Differential Equation

Equating the two expressions for $i_C(t)$:

$$-C\,\frac{dV_{\rm out}(t)}{dt} = \frac{V_{\rm in}(t)}{R_1}$$

Rearranging:

$$\frac{dV_{\rm out}(t)}{dt} = -\frac{V_{\rm in}(t)}{R_1 C}$$

This is the governing differential equation of the inverting integrator. The output voltage changes at a rate proportional to the input voltage, with a sign inversion and scaling by $\frac{1}{R_1 C}$.

Step 4: Integration to Get Output Formula

Integrating both sides of the differential equation from 0 to t:

$$\int_0^t \frac{dV_{\mathrm{out}}(\tau)}{d\tau}\, d\tau = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

The left side evaluates directly:

$$V_{\mathrm{out}}(t) - V_{\mathrm{out}}(0) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

Assuming the capacitor is initially uncharged, $V_{\mathrm{out}}(0) = 0$:

$$\boxed{V_{\mathrm{out}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau}$$

This is the **inverting integrator output formula**.

Circuit Description

The circuit has two stages:

1. **Stage 1 – Inverting Integrator**: input resistor $R_1 = 10\,\mathrm{k\Omega}$, feedback capacitor $C\,\mu\mathrm{F}$. Converts the square wave to a triangular ramp.
2. **Stage 2 – Inverting Amplifier**: input resistor $1\,\mathrm{k\Omega}$, feedback resistor $R\,\mathrm{k\Omega}$. Inverts and scales the triangular ramp.

The transfer functions are:

$$V_{\mathrm{int}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

$$V_{\mathrm{out}} = -\frac{R\,[\mathrm{k\Omega}]}{1\,[\mathrm{k\Omega}]}\, V_{\mathrm{int}} = -R\ V_{\mathrm{int}}$$

Input Signal

The input is a bipolar square wave of amplitude $V_p = 5\,\mathrm{V}$:

$$V_{\mathrm{in}}(t) = \begin{cases} +5\,\mathrm{V} & 0 \leq t < T/2 \\ -5\,\mathrm{V} & T/2 \leq t < T \end{cases}$$

The time parameters are:

1. Frequency: $f = 50\,\mathrm{Hz}$
2. Period: $T = 1/f = 20\,\mathrm{ms}$
3. Half-period: $T/2 = 10\,\mathrm{ms} = 0.01\,\mathrm{s}$
4. Amplitude: $V_p = 5\,\mathrm{V}$

A bipolar square wave integrated over each half-period produces a linear ramp. Alternating positive and negative half-cycles produce a triangular wave at Stage 1 output.

Stage 1: Integrator Analysis

The ramp magnitude over one half-period is:

$$\Delta V_{\mathrm{int}} = \frac{V_p \cdot (T/2)}{R_1 \cdot C}$$

Substituting $V_p = 5\,\mathrm{V},\ T/2 = 0.01\,\mathrm{s},\ R_1 = 10^4\,\Omega,\ C_1 = C \times 10^{-6}\,\mathrm{F}$:

$$\Delta V_{\mathrm{int}} = \frac{5 \times 0.01}{10^4 \times C \times 10^{-6}} = \frac{0.05}{0.01\,C} = \frac{5}{C}\ \mathrm{V}$$

Since the $-5\,\mathrm{V}$ half-cycle produces an equal upward ramp, the peak-to-peak at Stage 1 output is:

$$\Delta V_{\mathrm{int,pp}} = \frac{5}{C}\ \mathrm{V}$$

Stage 2: Amplifier

Stage 2 gain magnitude:

$$|A_v| = \frac{R\,[\mathrm{k}\Omega]}{1\,[\mathrm{k}\Omega]} = R$$

Peak-to-peak at Stage 2 output:

$$V_{\mathrm{out,pp}} = R \times \frac{5}{C} = \frac{5R}{C}$$

Setting $V_{\mathrm{out,pp}} = 5\,\mathrm{V}$:

$$\frac{5R}{C} = 5 \quad \Longrightarrow \quad \frac{R}{C} = 1$$

Plot of Waveforms

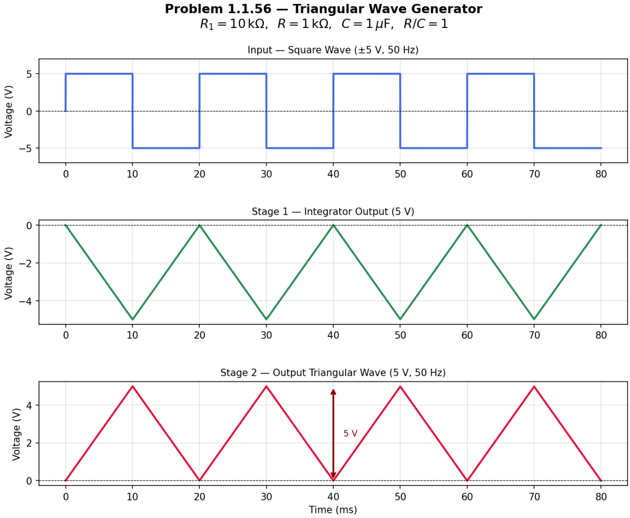

Fig : Waveforms

Python Code: Verification

```
import numpy as np

#  Given
Vin_amp = 5.0 # Bipolar square wave amplitude (V), i.e. 5 V
f = 50 # Hz
T = 1.0 / f # Period = 20 ms
R1 = 10**4 # Integrator input resistor = 10 k ()
R_in2 = 10**3 # Stage-2 input resistor = 1 k ()



#  Analysis
# During the +5V half-cycle, integrator ramps DOWN by delta.
# During the 5V half-cycle, integrator ramps UP by delta.
# Net peak-to-peak at Stage-1 output = delta (one half-swing).

r_c = (5*2*R1)/(Vin_amp*T)

r_c = r_c / 10**6

print(f"The value of R/C = {r_c}")
```

Python Code: Waveform Simulation

```
"""
plot_waveforms_1156.py  Problem 1.1.56
Simulate and plot waveforms for the triangular wave generator.
Condition: RC = 1 (R = 1 k, C = 1 F)
"""
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

#  Parameters
Vin_amp = 5.0 # Bipolar square wave amplitude (V)
f = 50 # Hz
T = 1 / f # Period = 20 ms
R1 = 10e3 # Integrator input resistor () = 10 k
C = 1e-6 # Capacitor (F) = 1 F
R2 = 1e3 # Stage-2 feedback resistor () = 1 k
R_in2 = 1e3 # Stage-2 input resistor () = 1 k
gain2 = R2 / R_in2 # = 1

dt = T / 2000
t = np.arange(0, 4 * T, dt)

#  Input: bipolar square wave 5 V
Vin = Vin_amp * np.sign(np.sin(2 * np.pi * f * t))

#  Stage 1: Inverting integrator (Euler)
Vint = np.zeros(len(t))
for i in range(1, len(t)):
    Vint[i] = Vint[i - 1] - (Vin[i - 1] / (R1 * C)) * dt
```

Python Code: Waveform Simulation

```
# Stage 2: Inverting amplifier
Vout = -gain2 * Vint

# Measure pp after settling
mask = t > T
Vpp_out = Vout[mask].max() - Vout[mask].min()
Vpp_int = Vint[mask].max() - Vint[mask].min()

# Plot
fig = plt.figure(figsize=(11, 8))
fig.suptitle(
    "Problem␣1.1.56␣␣Triangular␣Wave␣Generator\n"
    r"$R_1␣=␣10\,\mathrm{k\Omega},\;\;R␣=␣1\,\mathrm{k\Omega},\;\;"
    r"C␣=␣1\,\mu\mathrm{F},\;\;R/C␣=␣1$",
    fontsize=13, fontweight=’bold’
)
gs = gridspec.GridSpec(3, 1, hspace=0.55)
t_ms = t * 1e3

# Panel 1: Input
ax1 = fig.add_subplot(gs[0])
ax1.plot(t_ms, Vin, color=’royalblue’, lw=2.0)
ax1.set_title(’Input␣␣Square␣Wave␣(5␣V,␣50␣Hz)’, fontsize=10)
ax1.set_ylabel(’Voltage␣(V)’)
ax1.set_ylim(-7, 7)
ax1.axhline(0, color=’k’, lw=0.6, ls=’--’)
ax1.grid(True, alpha=0.35)
```

Python Code: Waveform Simulation

```python
#  Panel 2: Integrator output
ax2 = fig.add_subplot(gs[1])
ax2.plot(t_ms, Vint, color='seagreen', lw=2.0)
ax2.set_title(
    f'Stage 1  Integrator Output (5 V)',
    fontsize=10
)
ax2.set_ylabel('Voltage (V)')
ax2.axhline(0, color='k', lw=0.6, ls='--')
ax2.grid(True, alpha=0.35)

#  Panel 3: Final output
ax3 = fig.add_subplot(gs[2])
ax3.plot(t_ms, Vout, color='crimson', lw=2.0)
ax3.set_title(
    f'Stage 2  Output Triangular Wave (5 V, 50 Hz)',
    fontsize=10
)
ax3.set_ylabel('Voltage (V)')
ax3.set_xlabel('Time (ms)')
ax3.axhline(0, color='k', lw=0.6, ls='--')
ax3.grid(True, alpha=0.35)
# Annotate peak-to-peak arrow
y_max = Vout[mask].max()
y_min = Vout[mask].min()
x_ann = t_ms[mask][len(t_ms[mask])//3]
ax3.annotate('', xy=(x_ann, y_min), xytext=(x_ann, y_max),arrowprops=dict(arrowstyle='<->', color='darkred'
      , lw=1.8))
ax3.text(x_ann + 1.5, (y_max + y_min) / 2,f'5 V', color='darkred', fontsize=9, va='center')
plt.savefig('waveforms.png', dpi=150, bbox_inches='tight')
plt.show()
```